

Table of contents

Mutual Information	2
Question	2
Definition of Mutual Information	2
Formal Definition	2
Intuitive Interpretation	2
Concrete Example	3
The Venn Diagram: Visual Representation	4
Obvious Geometric Relations	5
Expressions of Mutual Information via Entropies	5
1. In terms of simple and conditional entropy	5
2. Symmetric expression	5
3. In terms of joint and marginal entropies	6
4. Expression with conditional entropies	6
Expressions via Relative Entropy (Kullback-Leibler Divergence)	6
1. Mutual information as KL divergence	6
2. Conditional form as expectation of KL divergences	7
3. Alternative symmetric form	7
Summary Table of Expressions	7
Illustrative Numerical Example	8
Situation: Weather forecast and ice cream sales	8
Calculations:	8
Visualization of Relations	9
Special Cases and Limits	9
1. Independent Variables	9
2. Deterministic Variables	10
3. Relation with Correlation	10
Practical Applications	10
1. Feature Selection	10
2. Representation Learning	10
3. Neuroscience	10
Pedagogical Synthesis	11
The Conversation Analogy	11
Mnemonic Formulas	11
Conclusion	11

Mutual Information

Question

Give the definition of mutual information.

Explain the Venn diagram.

Demonstrate different expressions for mutual information via entropies and relative entropy.

Definition of Mutual Information

Mutual information is a fundamental measure in information theory that quantifies the amount of information one random variable provides about another, and vice-versa. It is a measure of the statistical dependence between two variables.

Formal Definition

For two discrete random variables X and Y with joint distribution $p(x, y)$ and marginal distributions $p(x)$ and $p(y)$, mutual information is defined as:

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log_2 \frac{p(x, y)}{p(x)p(y)}$$

In expectation form:

$$I(X; Y) = \mathbb{E}_{p(x, y)} \left[\log_2 \frac{p(X, Y)}{p(X)p(Y)} \right]$$

For continuous variables:

$$I(X; Y) = \iint p(x, y) \log_2 \frac{p(x, y)}{p(x)p(y)} dx dy$$

Intuitive Interpretation

Mutual information answers the question: “**How much do we learn about X by observing Y (and vice-versa)?**”

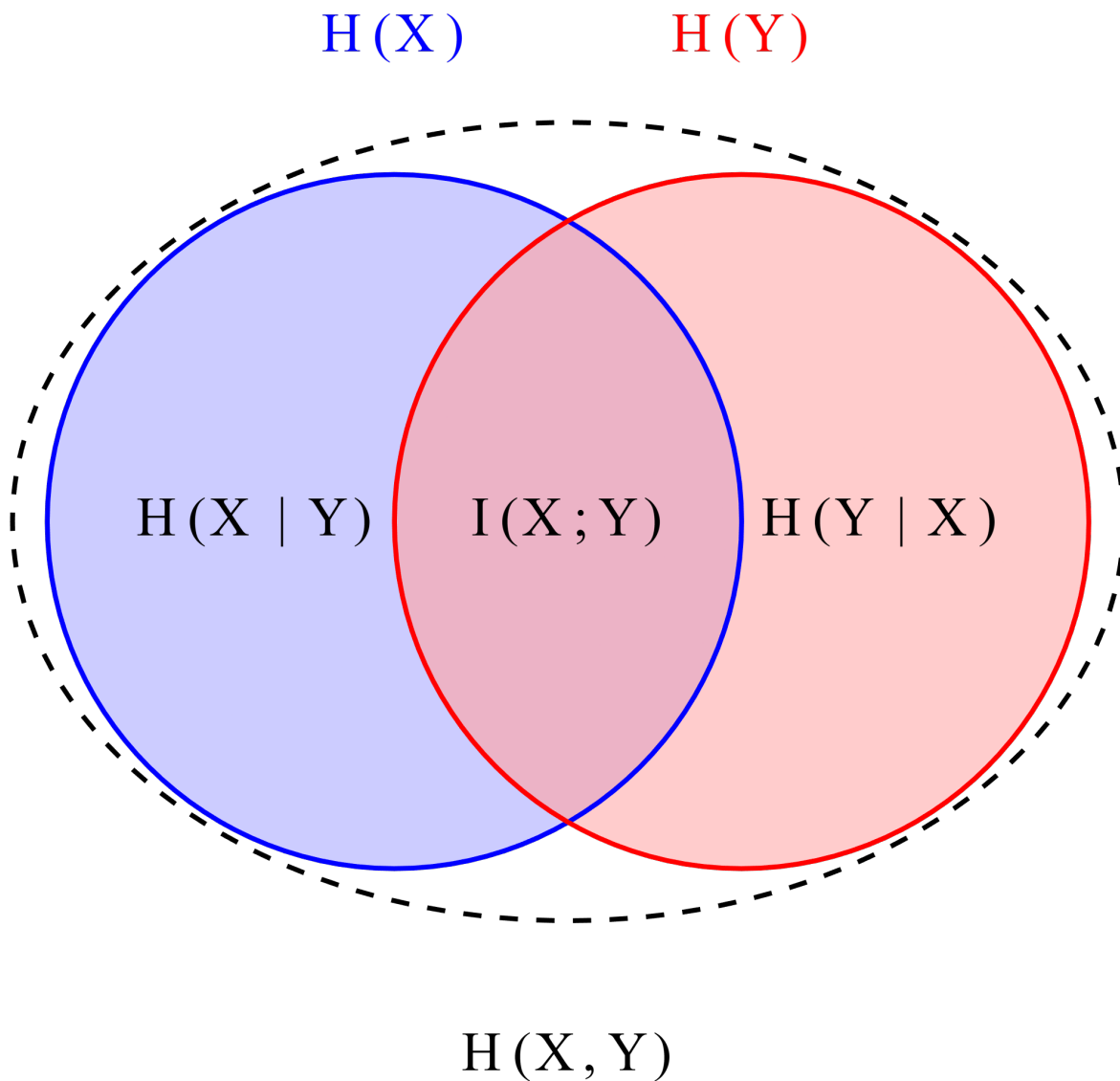
Concrete Example

Imagine two people communicating:

- $\mathbf{X}$: Message the first person wants to send
- $\mathbf{Y}$: Message the second person receives
- $\mathbf{I(X;Y)}$: Amount of information correctly transmitted

If the channel is perfect (no noise): $I(X;Y) = H(X)$ (everything is transmitted) If the channel is completely noisy: $I(X;Y) = 0$ (nothing is transmitted)

The Venn Diagram: Visual Representation



Legend and interpretation:

- $H(X)$: Entropy of X - measures total uncertainty about X
- $H(Y)$: Entropy of Y - measures total uncertainty about Y
- $H(X, Y)$: Joint entropy - total uncertainty of the pair (X, Y)
- $I(X; Y)$: Mutual information - information shared between X and Y

- $H(X|Y)$: Conditional entropy - remaining uncertainty about X when Y is known
- $H(Y|X)$: Conditional entropy - remaining uncertainty about Y when X is known

Obvious Geometric Relations

From the diagram:

1. $H(X, Y) = H(X) + H(Y) - I(X; Y)$
 - Union = sum of circles - intersection (to avoid double counting)
2. $H(X) = H(X|Y) + I(X; Y)$
 - Circle X = exclusive part + intersection
3. $H(Y) = H(Y|X) + I(X; Y)$
 - Circle Y = exclusive part + intersection

Expressions of Mutual Information via Entropies

1. In terms of simple and conditional entropy

$$I(X; Y) = H(X) - H(X|Y)$$

Interpretation: The information that Y provides about X is the reduction in uncertainty about X when Y is known.

Proof:

$$\begin{aligned}
 I(X; Y) &= \sum_{x,y} p(x, y) \log_2 \frac{p(x, y)}{p(x)p(y)} \\
 &= \sum_{x,y} p(x, y) \log_2 \frac{p(x|y)}{p(x)} \quad (\text{since } p(x, y) = p(x|y)p(y)) \\
 &= \sum_{x,y} p(x, y) \log_2 p(x|y) - \sum_{x,y} p(x, y) \log_2 p(x) \\
 &= -H(X|Y) + H(X) \quad (\text{by definition of entropies})
 \end{aligned}$$

2. Symmetric expression

$$I(X; Y) = H(Y) - H(Y|X)$$

Interpretation: Same as previous but symmetric - what X provides about Y.

3. In terms of joint and marginal entropies

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

Proof:

Starting from $I(X; Y) = H(X) - H(X|Y)$ and knowing that $H(X|Y) = H(X, Y) - H(Y)$, we have:

$$I(X; Y) = H(X) - [H(X, Y) - H(Y)] = H(X) + H(Y) - H(X, Y)$$

Interpretation: Mutual information is what “saves” bits when coding X and Y together rather than separately.

4. Expression with conditional entropies

$$I(X; Y) = H(X, Y) - H(X|Y) - H(Y|X)$$

Proof:

From $H(X, Y) = H(X) + H(Y) - I(X; Y)$, we can rearrange to get $I(X; Y) = H(X, Y) - [H(X, Y) - H(Y)] - [H(X, Y) - H(X)]$ which simplifies.

Expressions via Relative Entropy (Kullback-Leibler Divergence)

1. Mutual information as KL divergence

Fundamental relation:

$$I(X; Y) = D_{\text{KL}}(p(x, y) \parallel p(x)p(y))$$

Direct proof:

By definition of KL divergence:

$$\begin{aligned} D_{\text{KL}}(p(x, y) \parallel p(x)p(y)) &= \sum_{x, y} p(x, y) \log_2 \frac{p(x, y)}{p(x)p(y)} \\ &= I(X; Y) \quad (\text{exactly the definition of } I(X; Y)) \end{aligned}$$

Deep interpretation: Mutual information measures **the deviation from independence**.

- If X and Y are independent: $p(x, y) = p(x)p(y) \rightarrow \text{KL divergence} = 0 \rightarrow I(X; Y) = 0$
- If X and Y are dependent: $p(x, y) \neq p(x)p(y) \rightarrow \text{KL divergence} > 0 \rightarrow I(X; Y) > 0$

2. Conditional form as expectation of KL divergences

$$I(X; Y) = \mathbb{E}_{p(y)} [D_{\text{KL}}(p(x|y) \parallel p(x))]$$

Proof:

$$\begin{aligned} I(X; Y) &= \sum_{x,y} p(x, y) \log_2 \frac{p(x|y)}{p(x)} \quad (\text{since } \frac{p(x, y)}{p(x)p(y)} = \frac{p(x|y)}{p(x)}) \\ &= \sum_y p(y) \sum_x p(x|y) \log_2 \frac{p(x|y)}{p(x)} \\ &= \mathbb{E}_{p(y)} \left[\sum_x p(x|y) \log_2 \frac{p(x|y)}{p(x)} \right] \\ &= \mathbb{E}_{p(y)} [D_{\text{KL}}(p(x|y) \parallel p(x))] \end{aligned}$$

Interpretation: For each possible value of Y, we measure how much the conditional distribution of X given Y differs from the marginal distribution of X, then average over Y.

3. Alternative symmetric form

$$I(X; Y) = \mathbb{E}_{p(x)} [D_{\text{KL}}(p(y|x) \parallel p(y))]$$

Proof: Similar to previous by symmetry.

Summary Table of Expressions

Expression	Formula	Interpretation
Definition	$\sum p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$	Deviation from independence
Via entropies	$H(X) - H(X \parallel Y)$	Reduction in uncertainty about X by Y
Symmetric	$H(Y) - H(Y \parallel X)$	Reduction in uncertainty about Y by X
Joint-marginals	$H(X) + H(Y) - H(X, Y)$	Bit savings in joint coding
As KL	$D(p(x, y) \parallel p(x)p(y))$	Distance to independence
Conditional	$\mathbb{E}_{p(y)} [D(p(x y) \parallel p(x))]$	Average of conditional divergences

Illustrative Numerical Example

Situation: Weather forecast and ice cream sales

Let:

- **X**: Weather (0=Rain, 1=Sun)
- **Y**: Ice cream sales (0=Low, 1=High)

Distributions:

- $p(x = 0) = 0.3, p(x = 1) = 0.7$
- $p(y = 0) = 0.4, p(y = 1) = 0.6$
- Joint distribution:

	Y=0	Y=1
X=0	0.25	0.05
X=1	0.15	0.55

Calculations:

1. Independence check:

- If independent: $p(0, 0) = p(x = 0)p(y = 0) = 0.3 \times 0.4 = 0.12$
- But $p(0, 0) = 0.25 \rightarrow$ not independent

2. Entropies:

- $H(X) = -0.3 \log_2(0.3) - 0.7 \log_2(0.7) \approx 0.881$ bits
- $H(Y) = -0.4 \log_2(0.4) - 0.6 \log_2(0.6) \approx 0.971$ bits
- $H(X, Y) = -0.25 \log_2(0.25) - 0.05 \log_2(0.05) - 0.15 \log_2(0.15) - 0.55 \log_2(0.55) \approx 1.548$ bits

3. Mutual information:

- $I(X; Y) = H(X) + H(Y) - H(X, Y) \approx 0.881 + 0.971 - 1.548 = 0.304$ bits
- $I(X; Y) = D(p(x, y) \| p(x)p(y)) = 0.25 \log_2 \frac{0.25}{0.12} + 0.05 \log_2 \frac{0.05}{0.18} + 0.15 \log_2 \frac{0.15}{0.28} + 0.55 \log_2 \frac{0.55}{0.42} \approx 0.304$ bits

Interpretation: Knowing the weather provides about 0.3 bits of information about ice cream sales.

Visualization of Relations



Special Cases and Limits

1. Independent Variables

If $X \perp Y$, then:

- $p(x, y) = p(x)p(y)$
- $\log \frac{p(x, y)}{p(x)p(y)} = \log 1 = 0$
- Therefore $I(X; Y) = 0$

2. Deterministic Variables

If $Y = f(X)$ (deterministic function), then:

- $H(Y|X) = 0$ (knowing X , Y is certain)
- $I(X; Y) = H(Y)$ (all information about Y comes from X)
- But also $I(X; Y) \leq H(X)$

3. Relation with Correlation

Mutual information is more general than correlation:

- Correlation: measures linear relationships
- Mutual information: measures any dependence (linear or nonlinear)

Practical Applications

1. Feature Selection

In machine learning, we choose variables Y that maximize $I(X; Y)$ where X is the target variable.

Example: To predict house price (X), we choose among:

- Area (Y_1)
- Number of rooms (Y_2)
- Year of construction (Y_3)

We compute $I(X; Y_1)$, $I(X; Y_2)$, $I(X; Y_3)$ and keep the most informative ones.

2. Representation Learning

In autoencoders, we minimize $I(X; Z)$ where Z is the latent representation, to obtain a compact representation.

3. Neuroscience

Measure $I(S; R)$ where S is a stimulus and R is the neural response, to understand what information is encoded.

Pedagogical Synthesis

The Conversation Analogy

Imagine two friends talking:

- **H(X)**: Everything Pierre could say
- **H(Y)**: Everything Paul could hear
- **I(X;Y)**: What Paul actually understands from Pierre
- **H(X|Y)**: What Pierre says that Paul doesn't understand
- **H(Y|X)**: What Paul hears but doesn't come from Pierre (noise)

Mnemonic Formulas

1. “Shared information = what we know - what we ignore”

$$I(X;Y) = H(X) - H(X|Y)$$

2. “Together we are stronger, but we share”

$$H(X,Y) = H(X) + H(Y) - I(X;Y)$$

3. “Dependence is a distance to independence”

$$I(X;Y) = D(p(x,y) \parallel p(x)p(y))$$

Conclusion

Mutual information is a central concept that unites several notions from information theory:

1. **It is a measure of dependence** more powerful than correlation
2. **It elegantly relates entropies and KL divergence**
3. **Its visual interpretation** via Venn diagrams is intuitive

The Venn diagram clearly shows how mutual information is the **intersection** of information from X and Y, while the expressions via relative entropy reveal that it fundamentally measures **the gap between the joint reality and the independence hypothesis**.

This duality of perspectives makes mutual information an indispensable tool in data science, machine learning, neuroscience, and many other fields.